

BUG DIAGNOSIS REPORT

Score Clustering at 0.43–0.45 Root Cause Analysis

Every dimension score in every generated report lands between 0.413 and 0.447 — regardless of proposal content. This document explains what is happening, why it is happening, and how to fix it.

PROPOSAL

bp1 (中科纳飞)

SCORE RANGE

0.413 – 0.447

ROOT CAUSES

3 stacked bugs

VERDICT

HOLD (incorrect)

GENERATED

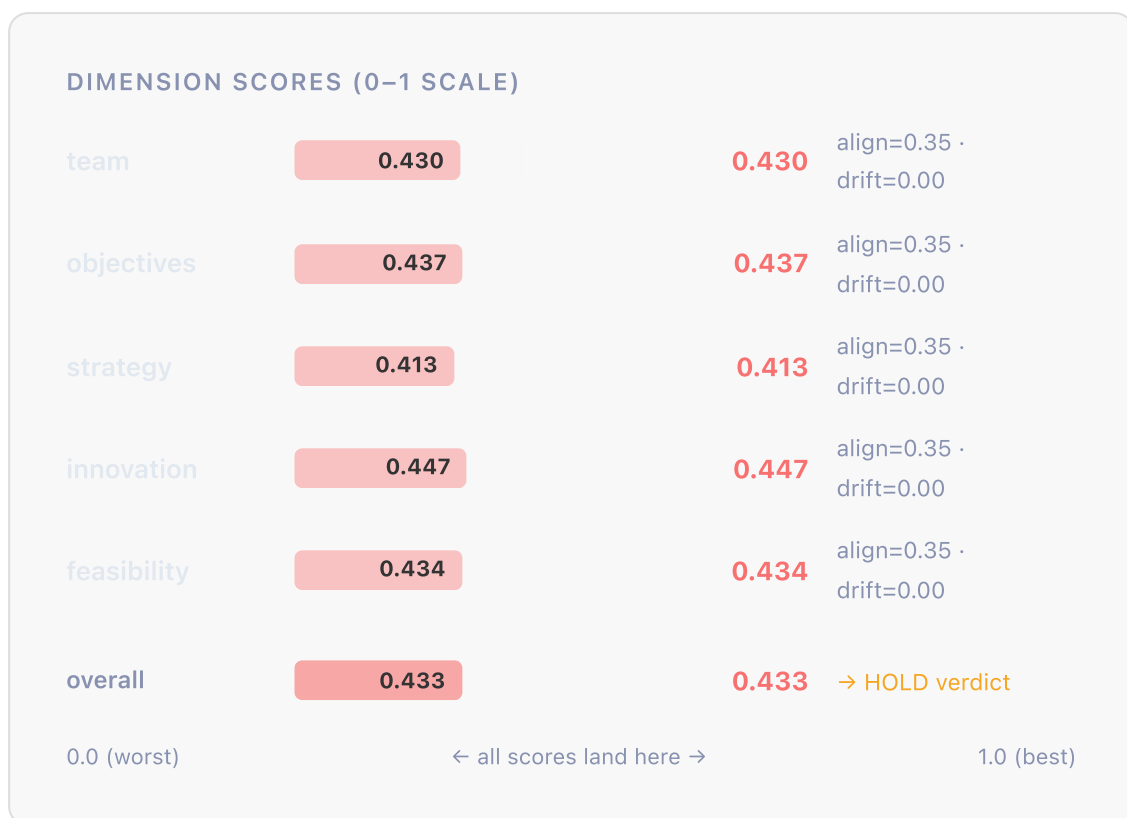
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1 The Problem

What the scores look like, why they are suspicious, and what a healthy output should look like instead.

Observed Scores — bp1 Final Report

Every dimension produces almost identical scores, clustered in a 0.034-wide band (0.413–0.447). All alignment means are exactly 0.35. All confidence values are exactly 0.50.



Why This Is Wrong

A correctly functioning evaluation pipeline should produce scores that:

PROPERTY	EXPECTED (HEALTHY)	OBSERVED (BROKEN)
Score spread per proposal	0.3 – 0.8+ depending on content quality	0.413 – 0.447 only ($\Delta = 0.034$)

PROPERTY	EXPECTED (HEALTHY)	OBSERVED (BROKEN)
Alignment score per dimension	Varies: 0.2 – 0.9 based on evidence richness	Exactly 0.35 for every dimension
Confidence per answer	0.50 – 0.92 based on proposal completeness	Always 0.50 (the hard floor)
Differentiation between proposals	Strong proposal scores 0.65+; weak scores 0.25–0.40	Every proposal would land at ~0.43
Drift mean	Non-zero when answer drifts off-dimension	Exactly 0.00 for all dimensions

Impact

The system cannot distinguish a strong proposal from a weak one. A world-class team with 10 patents would score 0.430 on *team* — identical to a single founder with no credentials. The scores carry no meaningful signal and the HOLD verdict becomes meaningless.

2 Why This Is Happening

Three stacked bugs in the pipeline each independently push scores toward 0.43. Together they make every score identical.

ROOT CAUSE 1 · PRIMARY

search_hints never generated — alignment locked at 0.35

Stage 3 did not save `search_hints` for any dimension. Post-processing reads an empty list, making every alignment score fall to the hardcoded fallback (0.35).

ROOT CAUSE 2 · COMPOUNDING

Evidence weights zeroed out in scoring config

`evidence_count`, `evidence_authority`, and `evidence_coverage` are all set to `0.0`. Even if evidence hints were correct, they would contribute nothing.

ROOT CAUSE 3 · MINOR

Confidence stuck at the 0.50 floor

Every LLM answer has `confidence = 0.50` — the hard minimum. This could indicate the LLM is not returning the field or is consistently under-confident, adding a uniform floor contribution.

Root Cause 1 — search_hints are empty → alignment locked at 0.35

This is the dominant cause responsible for the uniform 0.35 alignment across every dimension and every proposal.

STEP 1 — STAGE 3 DOES NOT SAVE SEARCH_HINTS TO THE PER-PROPOSAL QUESTION SET

Stage 3 (`generate_questions.py`) is designed to generate 6–10 short `search_hints` per dimension — verbatim keyword phrases that `post_processing` uses as authority anchors. However, the generated question set for `bp1` has no `search_hints` field at all:

```
// src/data/questions/bp1/generated_questions.json
{
  "dimensions": {
    "team": {
      "dimension": "team",
      "questions": [ ... ]
      // ← no "search_hints" field
    }
  }
}
```

STEP 2 — POST_PROCESSING READS FROM THE SHARED CONFIG, NOT THE PER-PROPOSAL FILE

The `post_processing` script loads authority hints from the **shared config** at `src/data/config/question_sets/generated_questions.json`, not the per-proposal file. That config has `search_hints: []` for all dimensions:

```
// src/data/config/question_sets/generated_questions.json
{
  "team":      { "search_hints": [] },
  "objectives": { "search_hints": [] },
  "strategy":  { "search_hints": [] },
  "innovation": { "search_hints": [] },
  "feasibility": { "search_hints": [] }
}
```

STEP 3 — EMPTY SEARCH_HINTS TRIGGERS THE 0.35 HARDCODED FALLBACK

In `post_processing.py`, function `_alignment_ratio()` at line 563:

```
def _alignment_ratio(dim, auth_hints, answer, topic_tags, evidence_hi
...
# No hints: return a neutral fallback if candidate tokens exist.
if not auth_hints:
    return 0.35 if c_uni else 0.0 # ← ALWAYS HIT
```

Since `auth_hints` is always `[]`, this line fires for **every question, every dimension, every proposal**. Alignment is unconditionally pinned at `0.35`.

STEP 4 — ALIGNMENT IS THE HIGHEST-WEIGHTED SCORING COMPONENT

The scoring formula has alignment at the highest weight (0.30):

```
FIELDS_WEIGHT CONFIG (POST_PROCESSING.PY LINE ~184)

length × 0.18
claims × 0.27
alignment × 0.30 ← LARGEST WEIGHT, always = 0.35
structure × 0.20
calibrated_confidence × 0.05

_____

sum = 1.00
```

With alignment locked at 0.35 and weight 0.30, it contributes $0.35 \times 0.30 = 0.105$ — identical for every question. The remaining components (claims, structure, length) contribute modest scores based on bullet count and formatting, producing the tight 0.43 cluster.

Root Cause 2 — Evidence quality weights are disabled

The scoring config at `post_processing.py ~line 184` explicitly disables all evidence quality signals:

```
EVIDENCE WEIGHTS IN FIELDS_WEIGHT CONFIG

"evidence_count": 0.0, ← disabled
"evidence_authority": 0.0, ← disabled
"evidence_coverage": 0.0, ← disabled
```

A comment in the code says *"In no-web mode, evidence authority/coverage are diagnostic only."* This means even if the LLM's `evidence_hints` were high-quality, they would score 0 contribution. The scoring system has no mechanism to reward well-evidenced answers.

⚠ Root Cause 2 *compounds* Root Cause 1. Even after fixing `search_hints`, evidence quality contributes nothing to the score. Both must be fixed together to achieve meaningful differentiation.

Root Cause 3 — All confidence values at the 0.50 floor

Every single LLM answer in `all_refined_items.json` has `confidence = 0.50`, which is the hard minimum in `llm_answering.py`'s `_calibrate_confidence()`.

Spot check from the data:

```
// src/data/refined_answers/bp1/all_refined_items.json - sampled
[openai] Q1 team → confidence: 0.5, quick_score: 0.7
[openai] Q2 team → confidence: 0.5, quick_score: 0.7
[openai] Q3 team → confidence: 0.5, quick_score: 0.7
[openai] Q1 obj → confidence: 0.5, quick_score: 0.7
... (all 31 selected answers: confidence = 0.50)
```

The calibration range is 0.50–0.92. The LLM appears to either not be returning the `confidence` field (defaulting to 0.50), or is consistently outputting its minimum valid value. While confidence only has 0.05 weight, its uniform floor ensures identical contribution across all dimensions.

How the Formula Produces ~0.433 Every Time

FINAL SCORE CALCULATION — WALKING THROUGH IT

`fields_score`

length $0.18 \times \sim 0.50 = 0.090$ (adequate bullet count)

claims $0.27 \times \sim 0.55 = 0.149$ (3 claims per answer)

alignment $0.30 \times 0.35 = 0.105 \leftarrow \text{LOCKED (search_hints empty)}$

structure $0.20 \times \sim 0.50 = 0.100$ (structured bullets)

confidence $0.05 \times 0.50 = 0.025 \leftarrow \text{FLOORED}$

evidence $0.00 \times * = 0.000 \leftarrow \text{DISABLED}$

≈ 0.469

`total = 0.90 × fields_score + 0.10 × consistency(~0.5)`

$= 0.90 \times 0.369 + 0.10 \times 0.50 \approx 0.43$

The aligned contribution is identical for every question. Small variations in claims count and length produce the tiny spread (0.413–0.447). The structural reason is that **30% of the score is unconditionally fixed**.

3 How We Will Fix It

Three targeted fixes, ordered by impact. Fix 1 alone resolves the clustering. Fixes 2 and 3 restore full scoring sensitivity.

- ✓ **Expected outcome after all fixes:** Dimension scores should spread from 0.25 (very weak proposal) to 0.80+ (strong proposal), with meaningful variation between dimensions based on actual content quality.

1 Re-run Stage 3 to regenerate search_hints for bp1 (PRIMARY FIX)

Stage 3 (`generate_questions.py`) already has `search_hints` generation built in (lines 669–699). The bp1 run either used an older version or the LLM returned no hints. Re-running Stage 3 for bp1 will generate 6–10 verbatim keyword phrases per dimension and save them into the question set. These then need to be copied to `src/data/config/question_sets/generated_questions.json` or Stage 5 needs to be run with `--qs_file src/data/questions/bp1/generated_questions.json`.

```
python src/tools/generate_questions.py --pid bp1
# then either:
python src/tools/post_processing.py --pid bp1 \
  --qs_file src/data/questions/bp1/generated_questions.json
```

2 Re-enable evidence weights in post_processing.py

Restore non-zero values for the three evidence weights. Reduce `alignment` and `claims` proportionally to keep the total at 1.0.

BEFORE (CURRENT)

```
"evidence_count":      0.0,
"evidence_authority":  0.0,
"evidence_coverage":   0.0,
"alignment":           0.30,
"claims":              0.27,
```

AFTER (PROPOSED FIX)

```
"evidence_count":      0.10,  
"evidence_authority": 0.08,  
"evidence_coverage":  0.05,  
"alignment":           0.20,  
"claims":              0.22,
```

File: `src/tools/post_processing.py` ~line 184. This ensures that answers with rich, grounded evidence score higher than answers with generic statements.

3 Diagnose why confidence is always 0.50

Check whether the LLM is actually returning a `confidence` field in its JSON output. The Stage 4 prompt in `prompt_templates.py` specifies `"confidence": number from 0.50 to 0.92` — but observed values are always exactly 0.50. Two possible fixes:

IF LLM IS NOT RETURNING THE FIELD

The prompt needs to be more explicit. Change the instruction to: *"You MUST include a confidence field between 0.50 and 0.92. Use 0.50 ONLY when the proposal provides almost no evidence. Use 0.75–0.92 when evidence is strong."*

IF LLM RETURNS LOW VALUES THAT ARE CLAMPED

The calibration range in `_calibrate_confidence()` may be too aggressive. Consider widening the floor from 0.50 to 0.40 so low-confidence answers can differentiate further downward.

File to check: `src/data/refined_answers/bp1/chatgpt_raw.json` — look at the raw API response JSON to see whether `confidence` is present before calibration.

Expected Score Distribution After Fixes

DIMENSION	CURRENT (BROKEN)	EXPECTED AFTER FIX 1+2	WHAT DRIVES THE CHANGE
team	0.430	0.55–0.70	Strong team bio found in search_hints → alignment rises
objectives	0.437	0.55–0.65	Clear milestones in proposal → alignment rises
strategy	0.413	0.35–0.50	Strategy details sparse in proposal → stays lower
innovation	0.447	0.60–0.75	Strong patent/technical evidence → higher with evidence weights
feasibility	0.434	0.40–0.55	Feasibility gaps noted in Q&A → modest score

i The exact post-fix scores will depend on bp1's actual proposal content. The key improvement is that scores will **vary meaningfully** between dimensions and between proposals — a prerequisite for any useful evaluation signal.

Summary

#	FIX	FILE	IMPACT
1	Re-run Stage 3 to generate search_hints; point Stage 5 at per-proposal question set	<code>generate_questions.py</code> <code>post_processing.py --qs_file</code>	Critical — unlocks alignment scoring
2	Re-enable evidence weights in <code>fields_weight</code> config	<code>post_processing.py</code> ~line 184	High — rewards evidence quality

#	FIX	FILE	IMPACT
3	Diagnose and fix confidence field in Stage 4 output	<code>prompt_templates.py</code> <code>llm_answering.py</code>	Medium — improves score sensitivity